

DATABASES

Laboratory Work 1

Entity-Relationship Diagram — International Airport Database

The Airport Database manages the operational core of an international airport: passenger bookings, flight scheduling, airline and airport reference data, boarding, baggage handling, and passenger security checks. The model below identifies the entities of this system, their attributes and keys, the relationships that connect them, and the resulting entity-relationship diagram.

1. Entity Identification

- AIRPORT — Stores airport reference data.
- AIRLINE — Stores airline reference data.
- FLIGHT — Scheduled and actual flight records.
- PASSENGER — Passenger profile data.
- BOOKING — A passenger's reservation on a flight.
- BOOKING_CHANGE — History of changes made to a booking.
- BOARDING_PASS — Boarding pass issued for a booking.
- BAGGAGE — Baggage registered against a booking.
- BAGGAGE_CHECK — Result of a baggage inspection.
- SECURITY_CHECK — Result of a passenger security check.

2-3. Attributes, Keys and Data Types

Each entity below lists its attributes, primary key (PK), and foreign keys (FK) with their referenced table.

AIRPORT

AIRPORT		
PK	airport_id	INT
	airport_name	VARCHAR NOT NULL
	country	VARCHAR NOT NULL
	state	VARCHAR NOT NULL
	city	VARCHAR NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

AIRLINE

AIRLINE		
PK	airline_id	INT
	airline_code	VARCHAR UNIQUE, NOT NULL
	name	VARCHAR NOT NULL
	country	VARCHAR NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

FLIGHT

FLIGHT		
PK	flight_id	INT
FK	airline_id → AIRLINE	INT NOT NULL
FK	departure_airport_id → AIRPORT	INT NOT NULL
FK	arrival_airport_id → AIRPORT	INT NOT NULL
	departing_gate	VARCHAR
	arriving_gate	VARCHAR
	scheduled_departure_time	DATETIME NOT NULL

FLIGHT

	scheduled_arrival_time	DATETIME NOT NULL
	actual_departure_time	DATETIME
	actual_arrival_time	DATETIME
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

PASSENGER**PASSENGER**

PK	passenger_id	INT
	first_name	VARCHAR NOT NULL
	last_name	VARCHAR NOT NULL
	gender	VARCHAR
	date_of_birth	DATE NOT NULL
	country_of_citizenship	VARCHAR NOT NULL
	country_of_residence	VARCHAR
	passport_number	VARCHAR UNIQUE, NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

BOOKING**BOOKING**

PK	booking_id	INT
FK	flight_id → FLIGHT	INT NOT NULL
FK	passenger_id → PASSENGER	INT NOT NULL
	status	VARCHAR NOT NULL
	booking_platform	VARCHAR
	ticket_price	DECIMAL(10,2) NOT NULL
	created_at	TIMESTAMP NOT

BOOKING		
		NULL
	updated_at	TIMESTAMP NOT NULL

BOOKING_CHANGE

BOOKING_CHANGE		
PK	booking_change_id	INT
FK	booking_id → BOOKING	INT NOT NULL
	change_description	VARCHAR NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

BOARDING_PASS

BOARDING_PASS		
PK	boarding_pass_id	INT
FK	booking_id → BOOKING	INT NOT NULL
	seat	VARCHAR NOT NULL
	boarding_time	DATETIME NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

BAGGAGE

BAGGAGE		
PK	baggage_id	INT
FK	booking_id → BOOKING	INT NOT NULL
	weight_kg	DECIMAL(6,2) NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

BAGGAGE_CHECK

BAGGAGE_CHECK		
PK	baggage_checking_id	INT
FK	booking_id → BOOKING	INT NOT NULL
FK	passenger_id → PASSENGER	INT NOT NULL
	check_results	VARCHAR NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

SECURITY_CHECK

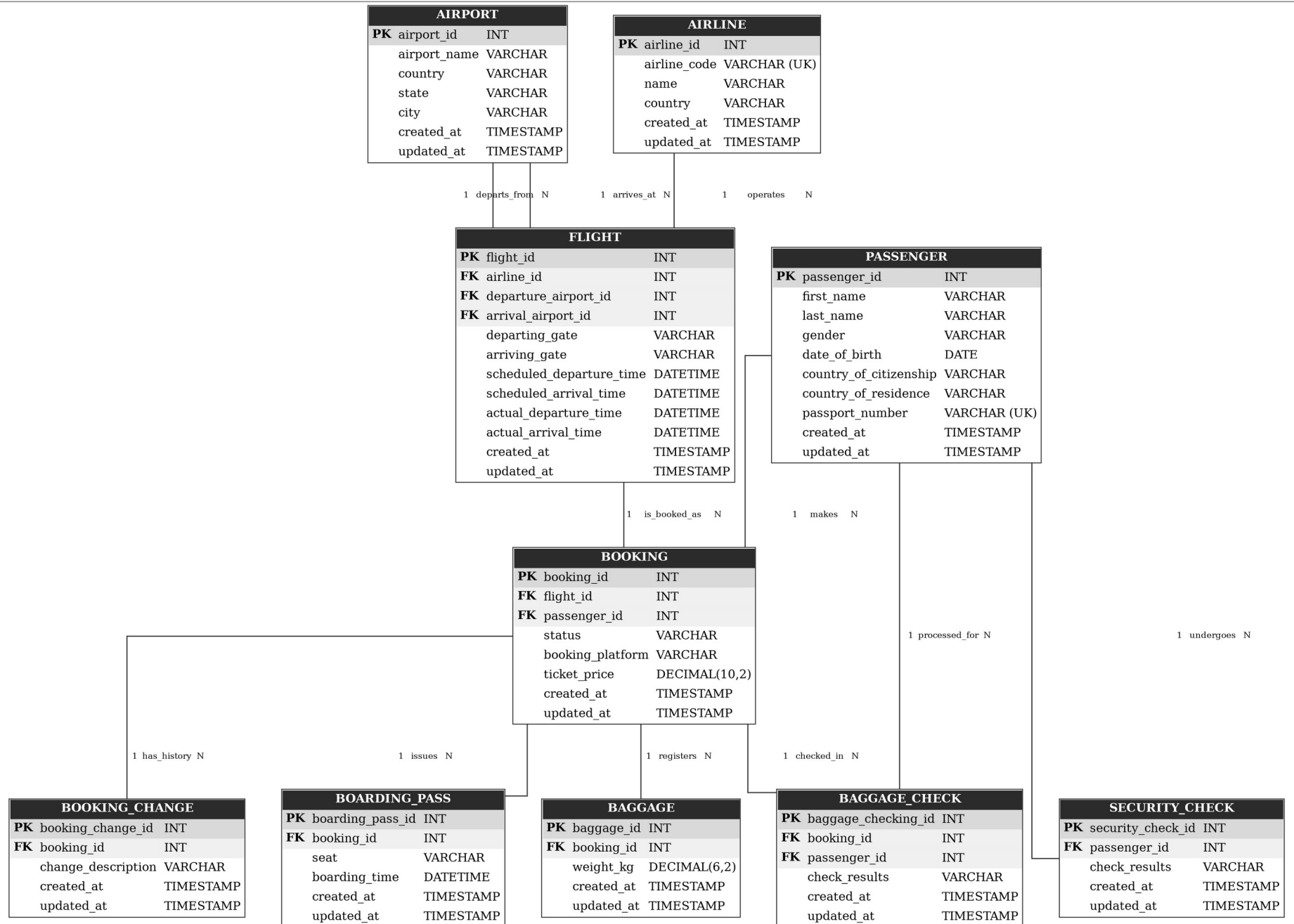
SECURITY_CHECK		
PK	security_check_id	INT
FK	passenger_id → PASSENGER	INT NOT NULL
	check_results	VARCHAR NOT NULL
	created_at	TIMESTAMP NOT NULL
	updated_at	TIMESTAMP NOT NULL

4. Relationships and Cardinality

Every relationship in this schema is one-to-many (1 : N), enforced through a foreign key on the “many” side referencing the primary key on the “one” side.

Entity A	Entity B	Cardinality	Relationship name
AIRLINE	FLIGHT	1 : N	operates
AIRPORT	FLIGHT (departure)	1 : N	departs_from
AIRPORT	FLIGHT (arrival)	1 : N	arrives_at
FLIGHT	BOOKING	1 : N	is_booked_as
PASSENGER	BOOKING	1 : N	makes
BOOKING	BOOKING_CHANGE	1 : N	has_history
BOOKING	BOARDING_PASS	1 : N	issues
BOOKING	BAGGAGE	1 : N	registers
BOOKING	BAGGAGE_CHECK	1 : N	checked_in
PASSENGER	BAGGAGE_CHECK	1 : N	processed_for
PASSENGER	SECURITY_CHECK	1 : N	undergoes

5-6. Entity-Relationship Diagram



7. Legend

- PK — primary key: unique identifier of a record.
- FK — foreign key: reference to the primary key of another entity.
- UK — alternate unique key (e.g. `passport_number`, `airline_code`).
- Dark header row — entity / table name.
- Shaded row (grey) — primary key attribute.
- Lightly shaded row — foreign key attribute.
- Connecting line, “1 ... N” — one-to-many relationship between the two entities.